HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and implemented a cross-platform **Rust** peer runtime for **Windows** and **Linux**, managing process lifecycles and background execution on a fleet of 5,000+ devices.
- Expanded the platform to **Android** devices using **NDK** and **JNI**, addressing foreground services, battery optimization, and mobile network constraints.
- Integrated native networking technologies, including **Chromium's network stack (Cronet)** and C++ libraries, into well-structured, reliable **Rust** libraries via **FFI**.
- Developed device-level observability and performance metrics, monitoring uptime, bandwidth consumption, and task success rates across distributed device fleets, reducing critical incident response time by **18%**.
- Collected and analyzed hardware and OS signals, including **TCP**, **TLS**, and **HTTP** fingerprints, to enhance network realism and support anti-detection engineering efforts.
- Participated in on-call rotations, investigated production incidents affecting **10,000+** devices, conducted post-mortems, and implemented improvements to system reliability.
- Containerized the peer runtime using **Docker** and deployed it to **AWS EC2** instances, managing configuration updates through **Jenkins** CI/CD pipelines.

SKILLS

Languages – Rust, C++, Java, Python, Kotlin, SQL

Platforms – Windows, Android, Linux, macOS

Networking – TCP, TLS, HTTP, Cronet, gRPC, mTLS

Development – FFI, JNI, NDK, Tokio, Tonic, Async Programming, Docker, Kubernetes

Observability – Prometheus, Grafana, OpenTelemetry, Splunk, Dynatrace

Cloud – AWS (EC2, S3, Lambda, CloudWatch)

PROJECTS

Cross-Platform Device Runtime (HCL Technologies)

- Developed a core **Rust** runtime for managing a distributed fleet of **Linux** and **Windows** devices, focusing on background execution and automated updates.
- Implemented **FFI** bindings to interface with native OS APIs, ensuring safe memory management and cross-language interoperability for critical system functions.
- Engineered robust error handling and logging mechanisms that improved system stability and reduced unhandled exceptions by **30%** across monitored devices.

Mobile Network Integration Service (HCL Technologies)

- Built a **Rust** library to integrate with **Android** foreground services and manage mobile network constraints, including traffic shaping and connection reliability.
- Utilized **JNI/NDK** to bridge **Rust** components with **Java/Kotlin** code, enabling seamless data exchange and execution within the Android environment.
- Designed and implemented network traffic analysis using **TCP** and **TLS** inspection, providing insights into device connectivity patterns for a team of 5 engineers.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)